

Network Penetration Testing Report

CloudExify Summer Internship 2026 — Cybersecurity Month 1, Project 1

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Task: Task 1 — Network Penetration Testing

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1. Objective

The objective of this project was to set up a safe, isolated virtual lab environment and perform an authorized penetration test against a deliberately vulnerable target machine, in order to practice the core stages of ethical hacking: reconnaissance, scanning, enumeration, and vulnerability identification.

2. Lab Environment Setup

A fully isolated virtual lab was built using VMware Workstation with two virtual machines:

- Attacker Machine: Kali Linux (contains penetration testing tools such as nmap, Metasploit, Wireshark)
- Target Machine: Metasploitable 2 (an intentionally vulnerable Linux server used strictly for training)

Both virtual machines were configured on the same NAT network so that they could communicate with each other while remaining isolated from the real internet and any outside systems.

Machine	Role	IP Address
Kali Linux	Attacker	192.168.193.129
Metasploitable 2	Target	192.168.193.128

Note on Ethics: All testing was performed exclusively on a private, self-owned virtual lab. No real, third-party, or production system was accessed at any point during this exercise, in line with the ethical hacking rules provided in the project guide.

3. Connectivity Test (Reconnaissance)

A ping test was run from Kali Linux to confirm that the target machine was reachable on the network:

Command: `ping 192.168.193.128`

Result: 5 packets transmitted, 5 packets received, 0% packet loss — confirming a successful and stable connection between the attacker and target machines.

4. Port Scanning (nmap)

An nmap scan was performed against the target to identify open ports and running services.

Command: `nmap 192.168.193.128`

The scan identified 23 open TCP ports out of 1000 scanned, indicating a large and intentionally exposed attack surface, which is expected for a deliberately vulnerable training target.

5. Service Version Enumeration

A version-detection scan was then run to identify the exact software and version running on each open port.

Command: `nmap -sV 192.168.193.128`

Port	Service	Version Detected
21	FTP	vsftpd 2.3.4
22	SSH	OpenSSH 4.7p1 (Debian)
23	Telnet	Linux telnetd
80	HTTP	Apache httpd 2.2.8
139/445	Samba	Samba smbd 3.X - 4.X
1524	Bindshell	Metasploitable root shell
3306	MySQL	MySQL 5.0.51a
5900	VNC	VNC protocol 3.3
6667	IRC	UnrealIRCd

6. Key Vulnerability Findings

6.1 vsftpd 2.3.4 — Backdoor Vulnerability

This specific version of vsftpd was compromised in 2011, when attackers modified the publicly distributed source code to include a hidden backdoor. Connecting with a specially crafted username can trigger a hidden remote shell, giving an attacker direct access to the system without a valid password.

6.2 UnrealIRCd — Backdoor Vulnerability

The IRC service running on port 6667 corresponds to a version of UnrealIRCd that was also distributed with a hidden backdoor in its source code, allowing remote attackers to execute commands on the server without authentication.

6.3 Samba smbd 3.X–4.X — Known Exploitable Version

The detected Samba version range includes builds affected by well-documented remote code execution vulnerabilities, which could allow an attacker to run commands on the target with elevated privileges.

6.4 Unencrypted Services (FTP, Telnet)

Both FTP (port 21) and Telnet (port 23) transmit data, including usernames and passwords, in plain text. Anyone capturing network traffic (for example with Wireshark) could read these credentials directly.

6.5 Pre-existing Root Shell (Port 1524)

Nmap identified port 1524 as an already-open root-level backdoor shell, requiring no authentication at all — the most critical finding in this assessment.

7. Recommendations

- Upgrade vsftpd, UnrealIRCd, and Samba to current, patched versions.
- Disable or replace unencrypted protocols (Telnet, FTP) with encrypted alternatives (SSH, SFTP).
- Close or firewall any ports and services that are not strictly required.
- Regularly scan production systems with nmap and vulnerability scanners to catch outdated or backdoored software early.
- Apply the principle of least privilege so that even if one service is compromised, an attacker cannot gain full system control.

8. Conclusion

This lab exercise successfully demonstrated the first stages of an ethical penetration test: building an isolated lab, confirming connectivity, scanning for open ports, and enumerating service versions to uncover multiple critical, real-world vulnerabilities. All testing was conducted in a fully authorized, isolated virtual environment in accordance with ethical hacking guidelines.

9. Appendix — Screenshots

The following screenshots document each stage of the lab setup and testing process.

Figure 1: Metasploitable 2 boot sequence completing successfully.

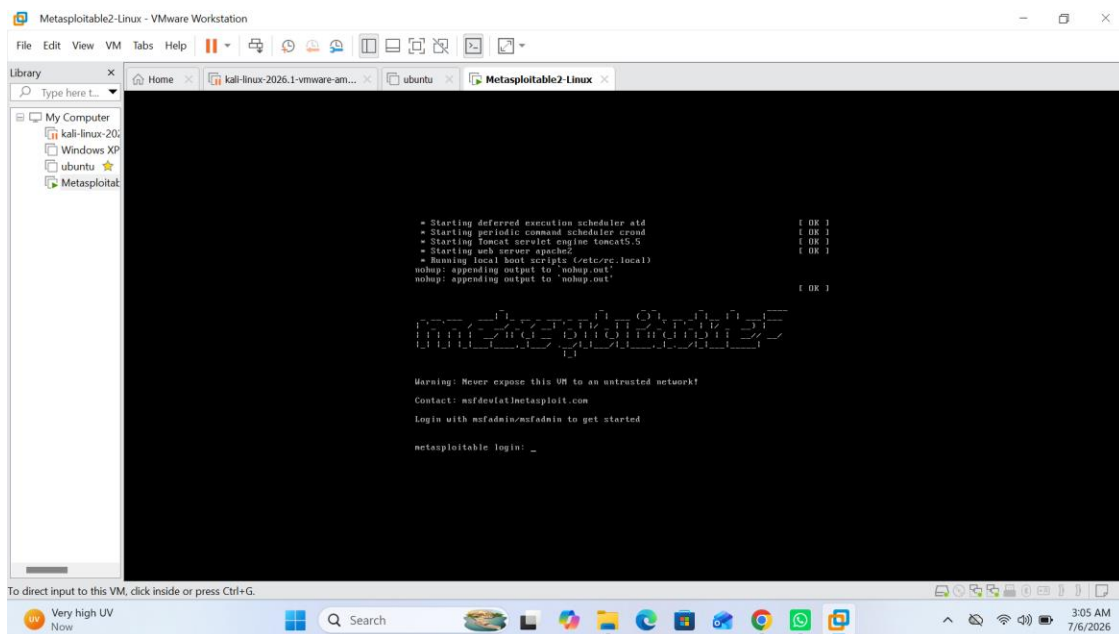


Figure 1 — Metasploitable booting up

Figure 2: Successful login to the Metasploitable target machine.

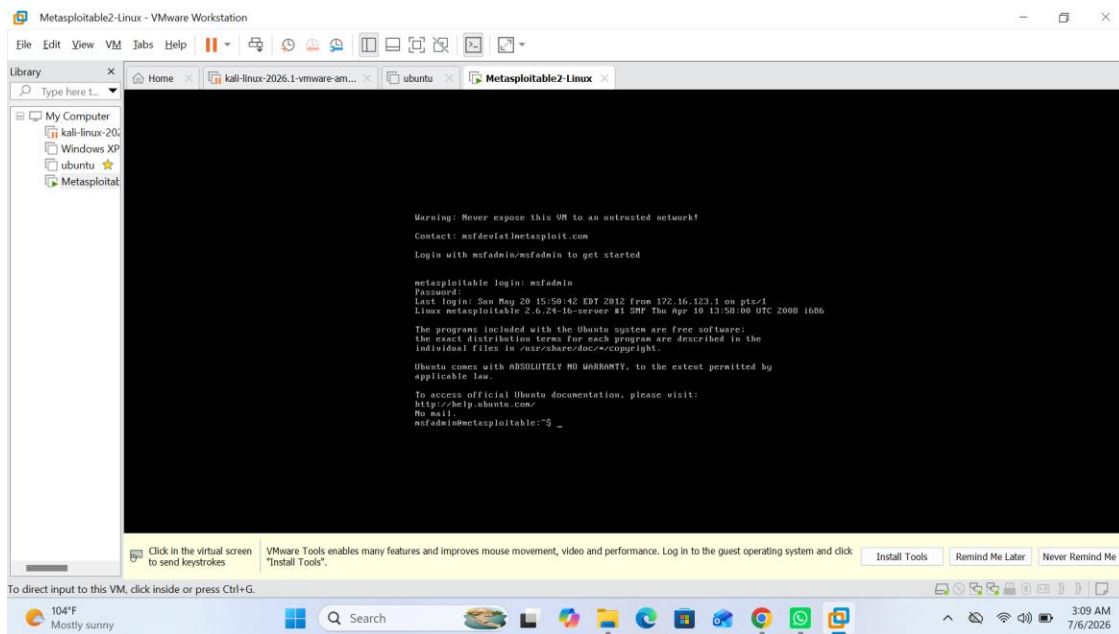


Figure 2 — Logged in to Metasploitable

Figure 3: Target IP address confirmed using ifconfig.

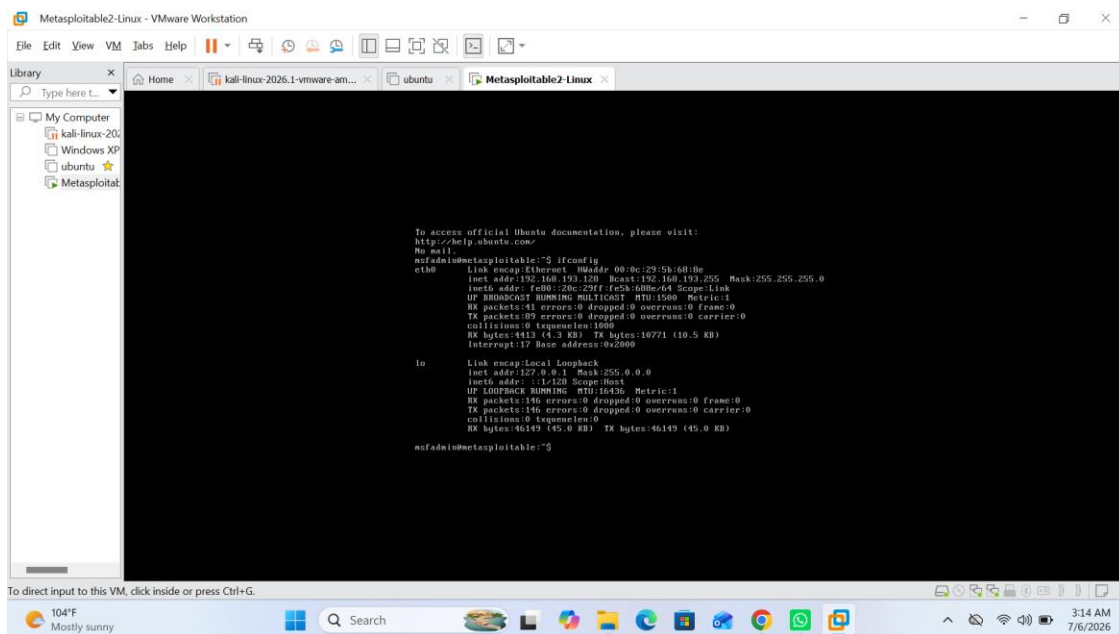


Figure 3 — ifconfig showing target IP 192.168.193.128

Figure 4: Network adapter reconfigured to NAT so both virtual machines share the same subnet.

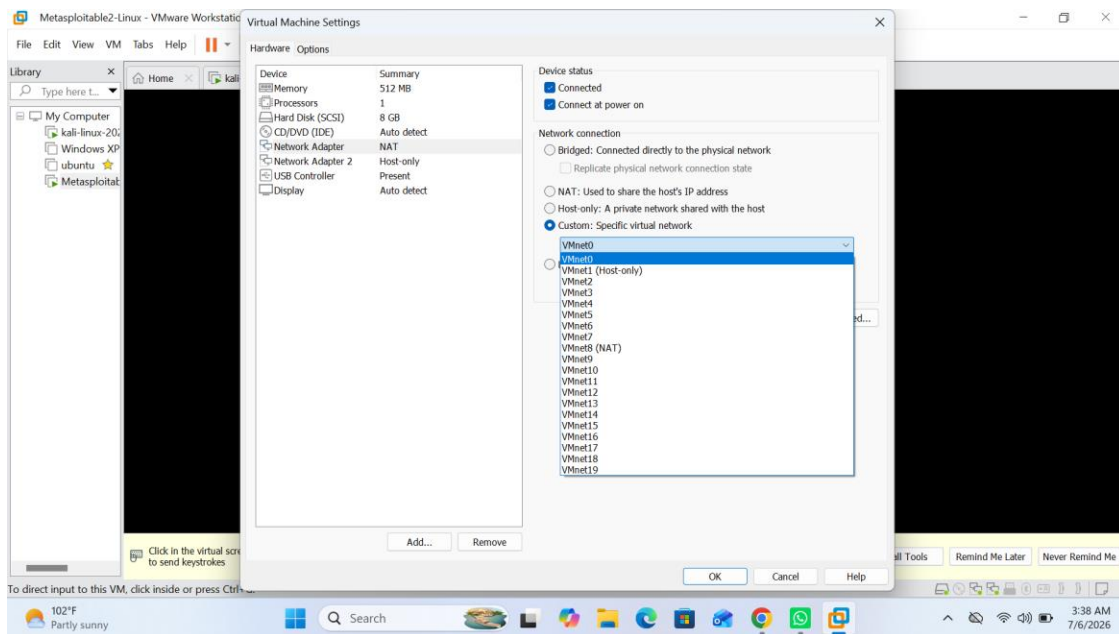


Figure 4 — Network adapter set to NAT

Figure 5: Wireshark actively capturing traffic on the eth0 interface during a Telnet session.

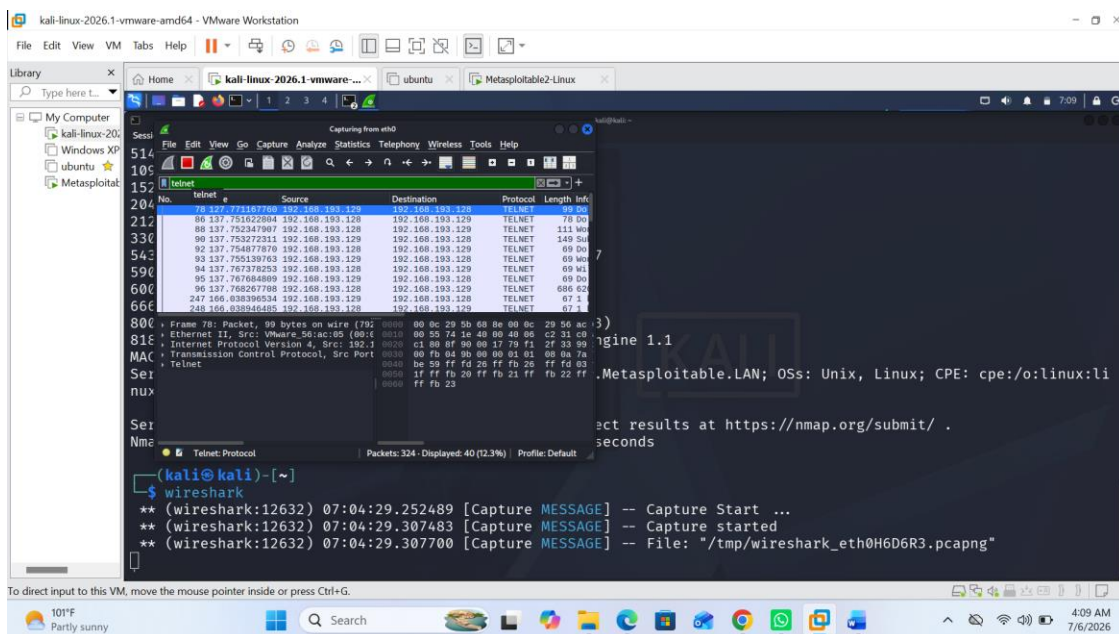


Figure 5 — Wireshark capturing Telnet traffic

Figure 6: Wireshark 'Follow TCP Stream' view reconstructing the Telnet session.

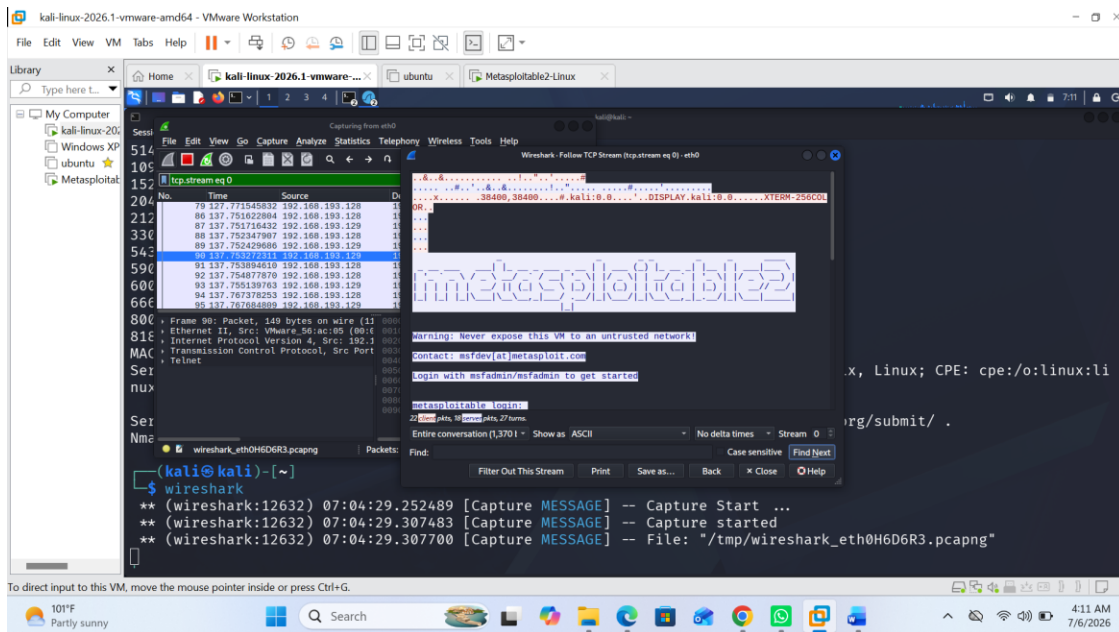


Figure 6 — Reconstructed Telnet session

Figure 7: Critical finding — the Telnet password ('msfdadmin') captured in plain, readable text.

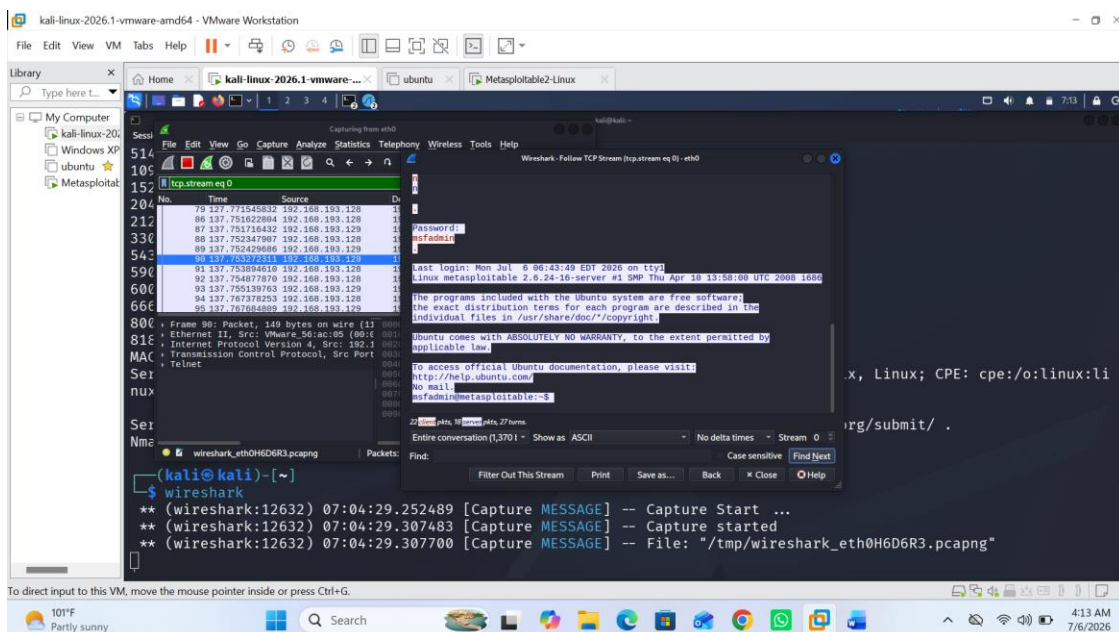


Figure 7 — Plain-text password exposed via Telnet

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